THE CORE PAPERS

Realization Calculus

Projectors, Transfer and Arithmetic

Komorebi P.

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A mathematical monograph in twelve chapters.

INTRODUCTION

Objects, operators and arithmetic

Realization Calculus studies how mathematical objects are generated from seeds, resolved into packets and related by operators. Projectors specify the components retained in a description. Transfer records their interaction and the passage from one resolution to another.

The first four chapters develop the Projector Resolved Seed Calculus. The argument then moves through multiresolution signals, evolution and memory, quantum dynamics and packet operator algorithms. The final chapters examine congruences, residual seeds, packet geometry and patching support.

Three questions recur. Which objects can be generated from the available seeds? Which relations survive a change of resolution? What additional hypotheses permit a compatible description to be realized in the intended analytic or arithmetic setting? Each construction is read with its own domain and assumptions.

The arithmetic conclusions retain their conditional character. Conjecture 66 in Chapter XI remains open. Chapter XII requires suitable residual and deformation data, patching and full support, a compatible Hecke point over the same coefficient field, classicality and an attached representation. A universal supply of these inputs is not a conclusion of the chapter.

The mathematical text and visual studies form the main sequence. The formal companions and glossary follow, with sources and references collected at the end.

READING MAP

Following the argument

The main sequence

Read Chapters I-XII in order for the development from seed constructions to conditional arithmetic realization. Each chapter opening connects its subject to the surrounding chapters and identifies the mathematical text and visual studies that follow.

Visual studies

The visual material stays with the chapter whose objects it illustrates. Figures and examples retain their stated inputs and ranges. A chapter identifier accompanies a global plate reference, as in Plate VIII.32.

Formal companions

The Lean guides are collected at the back. They identify the statements, hypotheses and source correspondences in their own formal scope. The machine-native treatment of Chapter I appears with its formal companion.

The back matter

Further studies retain their own hypotheses and open questions. The glossary gives short definitions with links to the relevant treatment. Sources and references are grouped by chapter and treatment at the end of the book.

Begin with Chapter I 1

Open the formal companions 1588

Open the glossary 2647

NOTATION AND REFERENCES

Conventions across the book

Local notation

Each chapter specifies its ambient spaces, scalar fields, domains and index sets. A symbol reused elsewhere need not denote the same object. Read every operator with its stated domain and codomain.

Finite and infinite settings

Finite algebra, limits and infinite-dimensional constructions have different assumptions. A formal identity does not by itself extend an operator to a larger space or supply continuity. The passage between settings is made where the text provides it.

Exact and numerical statements

Exact identities and numerical examples retain their stated roles. Numerical illustrations support the specified comparisons within their parameter ranges.

References within the volume

Continuous page numbers and PDF bookmarks navigate this book. Theorem, equation, figure and section numbers remain local to their named treatment unless a chapter prefix is given. Cite the chapter and treatment with the local number.

Citation labels

Bibliographies and source registers are collected at the end of the book. Their citation labels retain the numbering or short names used in the corresponding chapter and treatment. The accompanying electronic material preserves the supporting sources.

CONTENTS

Chapters I-VI

I Projector Resolved Seed Calculus I 1
Typed operator systems, seed rank, and exact packet coordinates

II Projector Resolved Seed Calculus II 98
Refinement towers, weighted transfer, and integral separation of operator packets

III Projector Resolved Seed Calculus III 177
Typed transfer, nonlinear packets, and obstruction theory

IV Projector Resolved Seed Calculus IV 279
Cyclic analytic functions, reciprocal families, and spectral zero data

V Multiresolution Signal Calculus 382
Exact coordinates, singular packets, and stable reconstruction

VI Transfer Semigroups and Memory 510
Probabilistic refinement, fractional operators, and exact hidden-state elimination

CONTENTS

Chapters VII-XII

VII Quantum Seeds and Effective Dynamics 659
Symmetry, geometric transport, quantum channels, and bounded refinement towers

VIII Packet Operator Algorithms 813
Exact compilation and GPU realization

IX Euler Congruences 930
Completion, factorial strengthening, and sharp dyadic geometry

X Universal Residual Seeds 1169
Deformation forests, transport categories, and conditional system completion

XI Tame and Wild Packet Geometry 1259
For local Langlands

XII Patching Support and Hecke Packets 1417
Conditional automorphic completion

CONTENTS

Further studies and reference

Further studies 1482

Arithmetic realization and its obstructions 1483

Visual studies of arithmetic realization 1551

Glossary 2647

Sources and references 2665

The scope of the companions

A written proof, a formal declaration and a finite computation answer different questions. The companions state their own coverage and the assumptions on which it depends.

Using the glossary

The glossary is a navigation aid. Its definitions are intentionally brief, and its references lead to the fuller statements and hypotheses in the text.

CONTENTS

Formal companions

I Projector Resolved Seed Calculus I 1588

II Projector Resolved Seed Calculus II 1721

III Projector Resolved Seed Calculus III 1768

IV Projector Resolved Seed Calculus IV 1815

V Multiresolution Signal Calculus 1936

VI Transfer Semigroups and Memory 1992

VII Quantum Seeds and Effective Dynamics 2054

VIII Packet Operator Algorithms 2132

IX Euler Congruences 2220

X Universal Residual Seeds 2350

XI Tame and Wild Packet Geometry 2384

XII Patching Support and Hecke Packets 2617

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